

Assessment of preferential flow processes in a forest-reclaimed lignitic mine soil by multicell sampling of drainage water and three tracers

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Abstract

Predictions of the long-term development of newly established forest-reclaimed ecosystems are required for land use planning of post-mining landscapes. The geological and mineralogical composition, the small-distance heterogeneity and inclined structures of the mine spoil's overburden sediment mixtures, as well as the physical and chemical properties of acidic and lignitic mine soil components, have raised questions about the water flow and solute transport processes in these soils. The objective of this study was to quantify preferential flow processes and spatially resolved solute transport in order to better distinguish between dominating processes for such soils. The experimental study was carried out by determining spatially resolved rates of throughfall, drainage and tracer leaching from underneath a 110 cm deep mine soil block. Drainage water was collected using 45 contiguous suction cells of 27 cm edge length. A tracer cocktail consisting of bromide, terbuthylazine, and deuterium was applied at the soil surface and subjected to natural infiltration. Tracer concentrations in drainage waters were analysed for a period of about 10 months. Then, the mine soil block was sampled completely in the form of 225 cubes of about 27 cm edge length to determine the residual bromide content. Tracer-labelled drainage in 110 cm depth occurred in spring 2001 over about 71% and in autumn 2001 over 35% of the total cross-sectional drainage area, respectively. Local drainage maxima shifted from spring to autumn, but also within drainage periods by 1–2 cell lengths. Bromide concentrations of drainage effluents varied by a factor of 4 between individual cells. Bypass-type preferential flow seems not to be a dominating transport mechanism since terbuthylazine in drainage waters occurred only singularly in spring with concentrations just above the detection limit. Perhaps the dilution in relatively large cells was too effective to detect modest contributions of tracer through preferential flow paths. Relative concentrations of deuterium and of bromide were akin during both drainage periods indicating similar transport behaviour of both tracers in the mine soil. The spatial distribution of residual bromide concentrations in the subsoil seemed to correspond with the inclined spoil ripple structures. The spatial patterns of leached bromide suggest a funnelling-type flow concentration. In this lignitic mine soil, preferential flow is presumably caused by the temporarily varying

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effects of multiple determinants, such as inclined dumping structures, spatial variability of hydraulic properties, water repellence, and the properties of lignitic dust and fragments with some immobile porosity.

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1. Introduction

Soil reclamation in mining as well as other areas is of increasing concern because of its potential long-term effects on ecological and hydrological systems. In the Lusatian mining district in eastern Germany alone, the post-mining landscapes cover an area of approximately 800 km² (Hüttl, 1998). Here, man-made soil systems, the so-called mine soils, develop on spoil heaps of overburden sediments. These sediments had been covering the lignite coal seams mostly in an aqueous environment below the groundwater level before they were drained, excavated and dumped in open-cast mining operations to form initially unsaturated spoil heaps. The overburden sediments frequently contain (among other components) Fe-sulphide minerals, such as pyrite and marcasite, which are subject to oxidative weathering when aerated. Oxidation leads to spoil acidification and to the generation of acid mine drainage (Evangelou, 1995). Such mine spoils have often been ameliorated by incorporating lime or flue ash into the upper layer of about 40–50 cm (Katzur and Haubold-Rosar, 1996). The reclaimed mine soils have mostly been afforested with pine tree species (Hüttl, 1998). Soil hydrological research at such sites aims at an improved understanding of soil dynamics and tree root development when attempting to estimate the long-term stability of the established forest ecosystems (e.g., Schaaf et al., 1999; Schaaf, 2001). Description of flow and transport in mine soils is also essential for evaluating the impact of the acid mine drainage on water quality (e.g., van Berk and Wisotzky, 1995; Evangelou, 1995) as well as effects of reclamation measures (von Willert and Stehouwer, 2003) on the quantity and quality of ground and surface water.

As a result of sediment dumping, the overburden mine spoils exhibit inclined-layered structures or ‘ripples’ and compacted impact zones (e.g., Matschak,

1969). In addition, a relatively large-scale (metre-scale) heterogeneity was suggested for zones of ‘pre-acidified’ sediments (Wisotzky, 1996). These are sediments that have been subject to oxidative weathering already at the outcrop side, for instance, at temporarily exposed open-pit surfaces. A relatively small-scale heterogeneity (centimetre-scale) of mine soil properties might result from mixing the overburden sediments of different geological origin during mining. Spatial heterogeneity of chemical and hydraulic properties of mine soils were reported for Lusatian (Prein and Mull, 1995) and other mine soils (e.g., Thum, 1974). Also, drainage was found to be spatially and temporally variable in mine soils (e.g., Katzur and Liebner, 1995). For relatively large spoil heap cross-sections, numerical simulation studies indicated that the above described spatial heterogeneities of spoil hydraulic and chemical properties have characteristic effects on acid mine drainage that cannot be neglected (e.g., Buczek et al., 2001; Gerke et al., 1998, 2001a).

Mine soil acidification may lead to phytotoxic element concentrations (Schaaf, 2001) that may adversely affect recultivation (Hüttl and Weber, 2001). Earlier quantitative studies of forest-reclaimed mine soil ecosystems (e.g., Scherzer, 2001; Wilden et al., 1999; Schaaf et al., 1999) estimated water and element budgets using standard assumptions (e.g., Darcian-based Richards’ equation and Fickian-based convection-dispersion equation) for describing flow and transport processes in mine soils. In contrast, however, a relatively large heterogeneity was observed at soil profiles, in matric potentials measured with tensiometers (Schaaf et al., 1999), in solute concentrations observed in extracted soil solutions (Gast et al., 2001), in water repellence (Gerke et al., 2001b), and, in particular, in flow paths observed in dye tracer experiments (Hangen et al., 2004). These findings gave rise to a demand for a better

understanding of preferential flow and transport processes in the reclaimed mine soil.

Preferential flow was assessed by means of 3D soil sampling (e.g., Heijs et al., 1996) and flow path visualisation (e.g. Flury et al., 1994; Hangen et al., 2002). Alternatively, preferential flow was quantified using tile drains (Lennartz et al., 1999) or lysimeters (Heuvelman and McInnes, 1997). The areas of lysimeter cells used range from relatively small (i.e., 10 cm²) in laboratory (Quisenberry et al., 1994) to large (i.e. 250 cm²) in field studies (e.g., Trapp et al., 1995). In another study, numerous suction cups were arranged upside down near the lysimeter bottom (Jene, 1998).

While in mostly fine-textured structured soils meso- and macropores act as preferential flow paths (i.e., bypass-type of preferential flow (Beven and Germann, 1982)), preferential flow can occur as ‘fingered’ flow (Glass et al., 1989) in coarse-textured non-structured soils or as ‘funnelling’ (Kung, 1990). Soil spatial heterogeneity can also lead to preferential pathways at larger spatial scales (Hendrickx and Flury, 2001). Preferential flow was studied for many different soils (e.g. see Bosch et al., 2001), however, only recently, macropore flow-type of preferential flow was identified on small spots of a non-vegetated black coal heap (Gueburt and Gardner, 2001). Also, heterogeneous and finger-type preferential flow patterns were observed in a lignitic mine soil plot under pine using a dye tracer irrigation experiment (Hangen et al., 2004).

Flow paths and transport processes in soil can further be characterized by using differently mobile tracers. Movement of reactive tracers along macropores over a relatively long vertical distance indicates rapid soil water flow and limited contact time for tracer sorption (Flury et al., 1995). Bromide may be used to indicate preferential flow if the tracer concentration peaks before the exchange of one pore volume of soil water (Lennartz et al., 1997). However, bromide is a semi-conservative tracer since it can be retarded by anion sorption (Gerritse and George, 1988) or accelerated by anion exclusion (Bowman, 1984). Isotopic components of the water molecule bear ideal transport properties and have been utilized as environmental tracers (e.g., Hangen et al., 2001a; Uhlenbrook et al., 2002). Artificial isotopic tracers, such as ¹⁵N, ¹⁸O and ²H, were used in irrigation

experiments (e.g., Priebe and Blackmer, 1989; Stamm et al., 2002).

In multiple tracer field-lysimeter studies, herbicide–bromide mixtures were applied for a rape–fallow–wheat rotation (Jene, 1998) and a spruce plantation (Trapp et al., 1995). Bromide and isotopic tracer mixtures were applied on agricultural sites (e.g., Schoen et al., 1999; Stamm et al., 2002) and on a non-vegetated lignitic mine soil (Knappe et al., 2001). However, only the herbicide–bromide tracers were collected by a spatially resolved lysimeter device.

The objective of this study was to assess preferential water movement and solute leaching in a forest-reclaimed lignitic mine soil. One aim was to investigate the spatial structure of flow paths to get better insight in processes of water movement in mine soils. A comparatively large soil block area should give relatively representative drainage and leaching rates. Spatially resolved tracer effluent should elucidate effects of tree root systems and mine spoil sediment structures. Differently adsorbing tracers should be tested to better differentiate transport properties. The investigations aim at a better characterisation of the leaching potential and the water availability for forest stands on mine soils and at providing hypotheses and data for subsequent model-based analyses.

2. Materials and methods

The field experiment was carried out at the reclaimed mine spoil heap ‘Bärenbrück’, which is located in the Lusatian lignite mining district north-east of the city of Cottbus in eastern Germany. The spoil was created in 1977 from overburden sediments of the nearby opencast lignite mines ‘Jänschwalde’ and ‘Cottbus-Nord’. At this site, mean annual precipitation for the period 1996–1999 amounted to 850 mm year⁻¹ (Scherzer, 2001), which ranged above the regional long term average of 573 mm year⁻¹ (Veit et al., 1987).

The experimental plot was selected according to a soil survey. It should be located in an area with lignitic and pyritic tertiary overburden sediments in order to be comparable with sites used for 1D water and element balance monitoring studies (Schaaf et al., 1999). The mine soil was described by



Fig. 1. Photo of mine soil profile of the 'Bärenbrück' test site showing ameliorated 40 cm topsoil, inclined dumping structures in the subsoil (i.e., spoil ripples), and lignitic fragments as black spots. The photo shows the front of the 4th grid-block row in y-direction. In the central section, a pine root stub is visible.

Neumann (1999) as an Anthropotic Regosol according to FAO-UNESCO (1978). Underneath an Ai horizon is a lyC horizon composed of sandy tertiary mine spoil material that contains about 10 vol% of lignitic components in form of dust and fragments of up to 5 cm in diameter (Neumann, 1999), which can be identified as dark spots in Fig. 1. Below a depth of about 40 cm, inclined dumping structures can be observed (Fig. 1). The predominant particle size fraction of the mineral component is fine sand (i.e., 0.063–0.2 mm equivalent diameter). The upper 40 cm were ameliorated with 190 metric tons of CaO ha⁻¹ in form of flue-ash that was incorporated into the mine soil by deep ploughing. The soil solution of the subsoil was highly acidic with pH-values around 2 (Wilden et al., 1999).

Pinus nigra was planted in 1982 with a row spacing of 120 cm at a density of 7375 trees ha⁻¹ (Fichter and Hüttel, 2000). The rooting depth of the pine trees next to our experimental site was mostly limited to the ameliorated upper 40 cm. At some spots, however, tree roots were found also in greater depths while other spots in the topsoil were free of roots (Hangen et al., 2001b). The field installations for measuring soil water balance components at the cell-lysimeter plot of 250 cm length and 135 cm width consisted of precipitation gauges, tensiometers, and multiple lysimeter cells (Fig. 2). The measurements were carried out over the period June 1st 1999 to September 21st 2001. The tracer experiment period was from November 7th 2000 to September 21st 2001.

Throughfall was measured by means of 18 plastic bulk samplers with cylindrical openings of 50 cm². The bottom parts of the samplers were put into small holes in the soil such that the collecting surface was about 10 cm above the surface of the litter layer. The samplers were arranged in three rows at a regular 42 cm-grid. Throughfall was collected twice a week. The temporal resolution of bulk precipitation rates including other meteorological data, such as air temperature, global radiation, wind speed, and relative humidity, was recorded at a nearby automated station outside the forested area. To obtain values for the depth of infiltrated water, the litter interception had to be considered. Water infiltration, Inf [L T⁻¹], was estimated by subtracting 15% of the calculated potential evapotranspiration, ET_{pot} [L T⁻¹], from the measured throughfall, P_{through} [L T⁻¹], according to Brechtel and Pavlov (1977) as:

$$\text{Inf} = P_{\text{through}} - 0.15\text{ET}_{\text{pot}} \quad (1)$$

and with ET_{pot} according to the Penman–Monteith combination equation (Monteith, 1980)

$$\text{ET}_{\text{pot}} = \frac{1}{L} \frac{s(R_n - G) + \frac{\rho c_p}{r_a}(E - e)}{s + \gamma \left(1 + \frac{r_s}{r_a}\right)} \quad (2)$$

where L is latent heat of vaporisation with dimensions [W kg⁻¹ d⁻¹], s is slope of saturated vapour pressure versus temperature curve [Pa °C⁻¹], R_n is net radiation [W m⁻²], G is heat flux density in soil [W m⁻²], ρ is density of air [g m⁻³], c_p is heat capacity of air at constant air pressure [J g⁻¹ °C⁻¹],

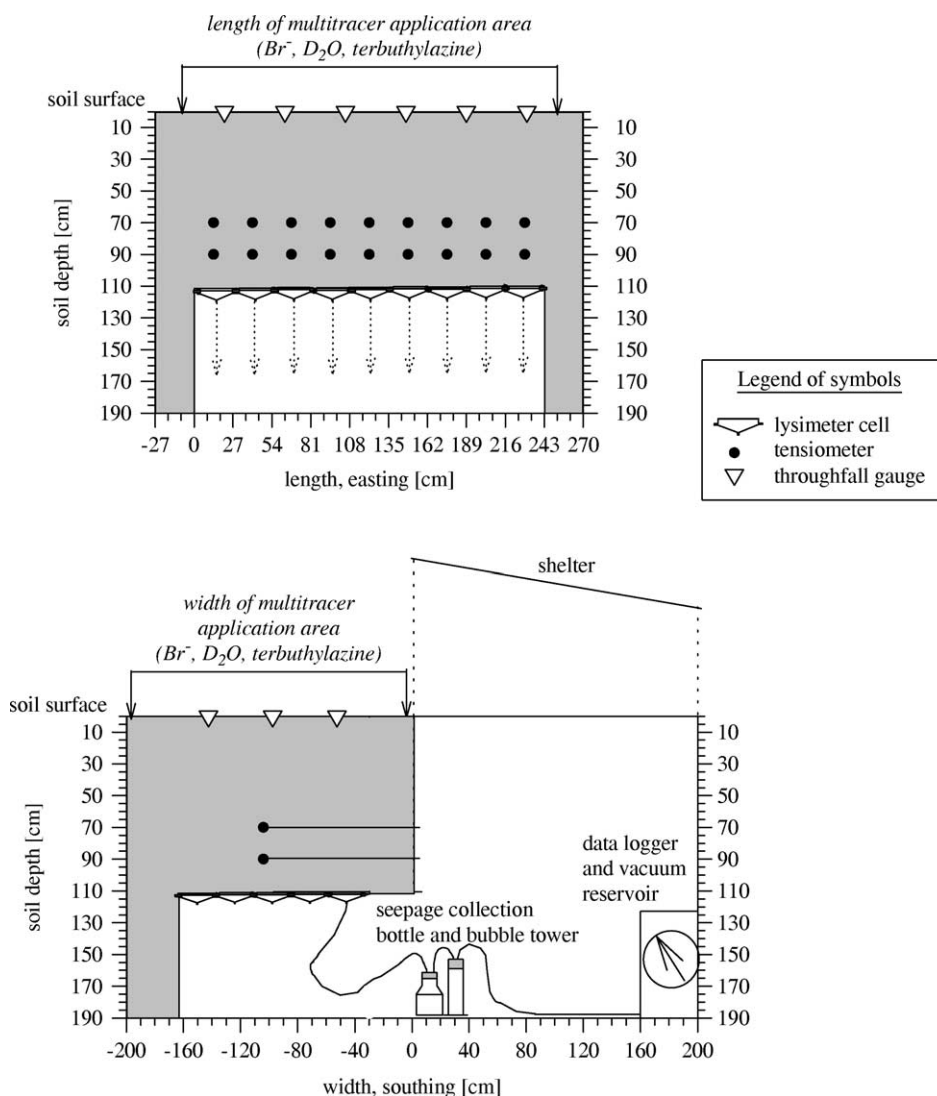


Fig. 2. Schematic vertical sections of the intact instrumented forest-reclaimed mine soil block (grey-coloured) at the 'Bärenbrück' field site: (a) front view (top); (b) side view (bottom). The figures show the tracer application area, the location of the throughfall collectors, horizontally installed tensiometers, and lysimeter cells. Arrows mark the flow direction of soil water of successively sampled lysimeter cells. The instruments were installed and operated from the neighbouring sheltered soil pit, where drainage water was withdrawn using bubble towers.

r_a is mean aerodynamic resistance of boundary layer of pine stand [$s\ m^{-1}$], $(E - e)$ is saturation deficit of vapour pressure curve $[-]$, γ is psychrometer constant [$Pa\ ^\circ C^{-1}$], and r_s is mean stomata resistance of pine stand [$s\ m^{-1}$]. Here, values of $r_a = 0.011\ s\ m^{-1}$ and $r_s = 200\ s\ m^{-1}$ during summer (May–October) and $r_s = 1000\ s\ m^{-1}$ during winter (November–April) half years were used.

Tensiometers were horizontally installed in 70 and 90 cm soil depth from an adjacent soil pit. At both soil depths, a row of 9 tensiometers were inserted such that the ceramic cells of 5 cm length and 2 cm diameter at their tips were located vertically above the centres of the middle rows of the suction cells, about 1 m apart from the soil pit (Fig. 2). We measured soil water suction relative to atmospheric pressure every 10 min

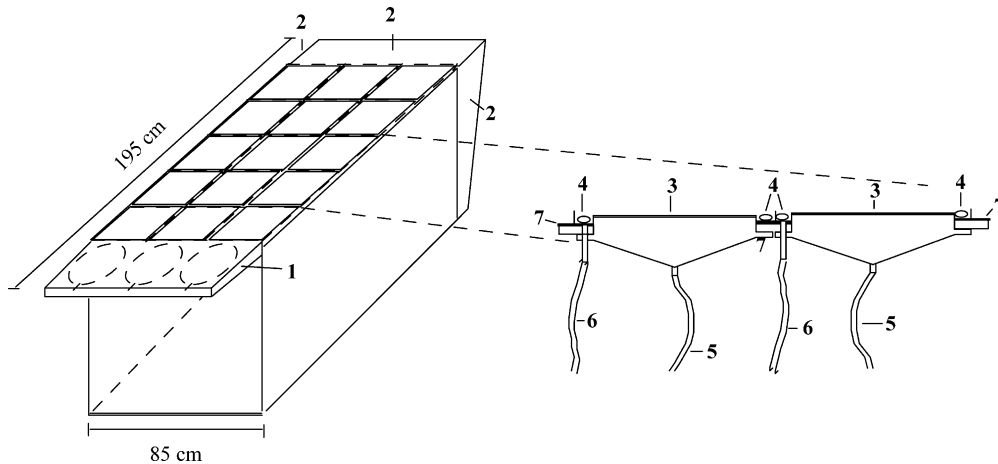


Fig. 3. Schematic drawing of one out of three steel frames, each of which carrying 15 individual lysimeter cells (left), and vertical cross-section of two adjacent lysimeter cells (right). The dotted lines show the spatial arrangement of perforated tubes for collection of non-capillary drainage. Numbers indicate: (1) front protruder; (2) cuttings edges; (3) nylon membrane supported by quartz-sand; (4) perforated tube (cross-sectional view); (5) tube for collecting suction cell drainage water; (6) tube for collecting non-capillary drainage water; (7) steel frame for cell attachment.

using pressure transducers (UPT 6; UP, Ltd., Cottbus, Germany) that were temperature-compensated and exhibited only minor drift. After calibration, the pressure transducers were mounted directly behind the ceramic cups inside the tensiometer shafts.

The lysimeter construction consisted of 3 steel frame segments (Fig. 3, left). Each segment had 15 squared openings of 27 cm edge length, cutting edges at the rear, and a protruding edge at the front. The segments were horizontally pushed into the mine soil from the adjacent soil pit using a hydraulic cylinder press such that the ceiling of all 3 segments was in 110 cm soil depth. The soil inside the segments was successively removed by hand in order to get access from the bottom to the squared openings, which were covered during installation. After removing the protective covers, the funnel-type suction cells were attached to the mine soil in 110 cm depth at all 45 openings (Fig. 3, right). Disturbed mine soil portions were carefully removed using a hand-brush. Voids between bottom of the soil and top of the funnel were filled with a water-suspension of quartz powder-sand, which had a texture comparable to that of the mine soil. The plastic funnels were covered with a nylon membrane of 20 μm mesh size (NY 20 HC; Hydro-Bios. Ltd. Kiel, Germany) that prevented an air-entry into the system below about 150 cm water column. The funnels were filled with quartz sand of 1 mm

particle diameter to support the membrane and were fully saturated with water before installation to ensure air-free suction lysimeter systems. The 45 funnel-type suction cells were intercepting a cross-sectional area of 3.3 m^2 . For collection of possibly occurring fast gravity drainage, perforated polyethylene pipes of 7 mm diameter were installed at the steel frames around each squared cell segment (Fig. 3, left). These drains should also prevent over-capillary water of one cell from moving laterally into neighbouring cells. To exclude boundary effects at the open side of the soil pit, the mine soil of the first 30 cm from the profile wall was not included in the experiment (Fig. 2b). Nevertheless, drainage pipes were installed at

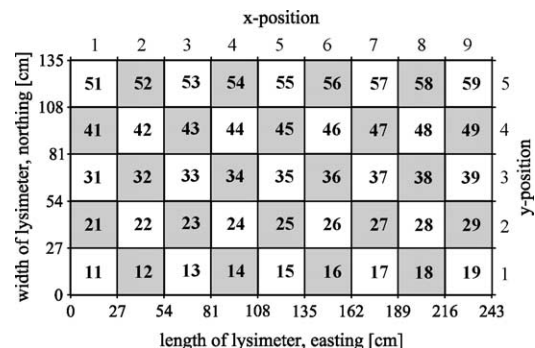


Fig. 4. Plan-view of the contiguous drainage area of three lysimeter segments installed side by side and identification numbers of lysimeter cells.

the protruding edges of the steel segments to remove surplus drainage water from the non-sampled 30 cm strip (Fig. 3, left). Note that the size of the squared cells of 27 cm edge length was selected according to observed finger-to-finger distances of a previous dye tracer experiment at the adjacent soil pit (Hangen et al., 2004). Identification numbers were designated to the individual lysimeter cells (Fig. 4).

The tracer cocktail was composed of potassium bromide (KBr) and terbuthylazine ($C_9H_{16}ClN_5$, as component of pesticide 'Gardoprim Plus'), which were both dissolved in 8 l of deuteriated water (D_2O). The concentrations were 3.125 g l^{-1} for Br^- and 0.0625 g l^{-1} for terbuthylazine, respectively. As recommended by Russow et al. (1996), water with an enrichment ratio of 90 atom.% deuterium ($\cong 900,000 \text{ ppm}$) was used. The deuterium concentration, δ^2H [‰], is expressed according to Moser and Rauert (1980) as the relative deviation from standard water of known deuterium signature $(^2H/^1H)_{V-SMOW}$, as:

$$\delta^2H = \frac{(^2H/^1H)_{\text{sample}} - (^2H/^1H)_{V-SMOW}}{(^2H/^1H)_{V-SMOW}} 1000 \quad (3)$$

where $(^2H/^1H)_{\text{sample}}$ is the relation between deuterium and hydrogen of the sample water and $(^2H/^1H)_{V-SMOW}$ is that of the Vienna-Standard Mean Ocean Water, respectively. The tracer cocktail solution was mixed such that a rate of 5 g m^{-2} bromide and of 0.1 g m^{-2} terbuthylazine was applied according to maximal values permitted in agriculture (LELF, 1999).

The tracer was applied on November 7th, 2000. The mean antecedent moisture content in the upper mineral soil (0–5 cm depth) on the day of application was about 12 vol% with a standard deviation of 1.9 vol% as determined using five 100 cm^3 soil cores sampled close to the experimental plot. The tracer solution was manually applied onto the litter-covered soil surface above the cell-lysimeter area by dripping technique using a multipette (Multipette 4780; Eppendorf Ltd., Hamburg, Germany) to ensure a homogeneous tracer distribution. For this, the total area was subdivided into $25 \times 25 \text{ cm}$ grids, each of which was further subdivided into squares of $2.5 \times 2.5 \text{ cm}$ (Fig. 5). Each 6.25 cm^2 area received 1 ml of tracer solution followed by 2.5 ml of distilled water in

order to wash the tracer solution from the litter downwards into the mineral soil. In total, 1.6 mm of tracer solution and 4 mm of distilled water were applied. The tracer application area was slightly larger ($2.64 \times 2 \text{ m}$) than that of the undermined soil area ($2.5 \times 1.35 \text{ m}$) in order to reduce effects of divergent lateral tracer movement on the drainage concentrations. After application, tracer movement in the mine soil block was left subject to naturally occurring infiltration events.

Drainage water was collected twice a week according to the following procedure: First, non-capillary gravity drainage water was collected, if present, from each cell by emptying the drain tubes. Second, cell drainage water was sampled from the tubes connected to the funnels by imposing specific suction. Values of the imposed pressure head, h (negative suction) at the cell surface in 110 cm depth, $h_{110} = h_{90}$, were determined from the actual tensiometer-measured suctions in 90 cm depth, h_{90} , by assuming a unit hydraulic gradient that implies a zero suction gradient:

$$\frac{\Delta(h+z)}{\Delta z} = 1 \quad (4)$$

where z is depth directed vertically upward. The funnel outflow tubes were closed again after drainage ceased (i.e. imposing a no-flow condition). The suction was created by a vacuum pump that was linked to a reservoir, and suction values were regulated by means of bubble towers (Fig. 2b). The drainage sampling procedure always started at the front left side towards the rear parts and moved row by row using three bubble towers at a time.

The collected drainage water samples were stored at 4°C and passed through a membrane filter of $0.45 \mu\text{m}$ mesh size before being analysed for bromide by means of ion-chromatography (Dionex 500 IC; Dionex Ltd., Sunnyvale, USA). Selected drainage water samples were analysed for terbuthylazine by High-Performance-Liquid-Chromatography (Gynkoteck UVD 160 S; Gynkoteck Ltd., Oregon City, USA) and for deuterium by mass spectrometry (IRMS Delta S; Finnigan MAT Ltd., Bremen, Germany). To consider the natural background input of deuterium, selected samples of throughfall, of rainfall and of snow cover, were analysed for deuterium.

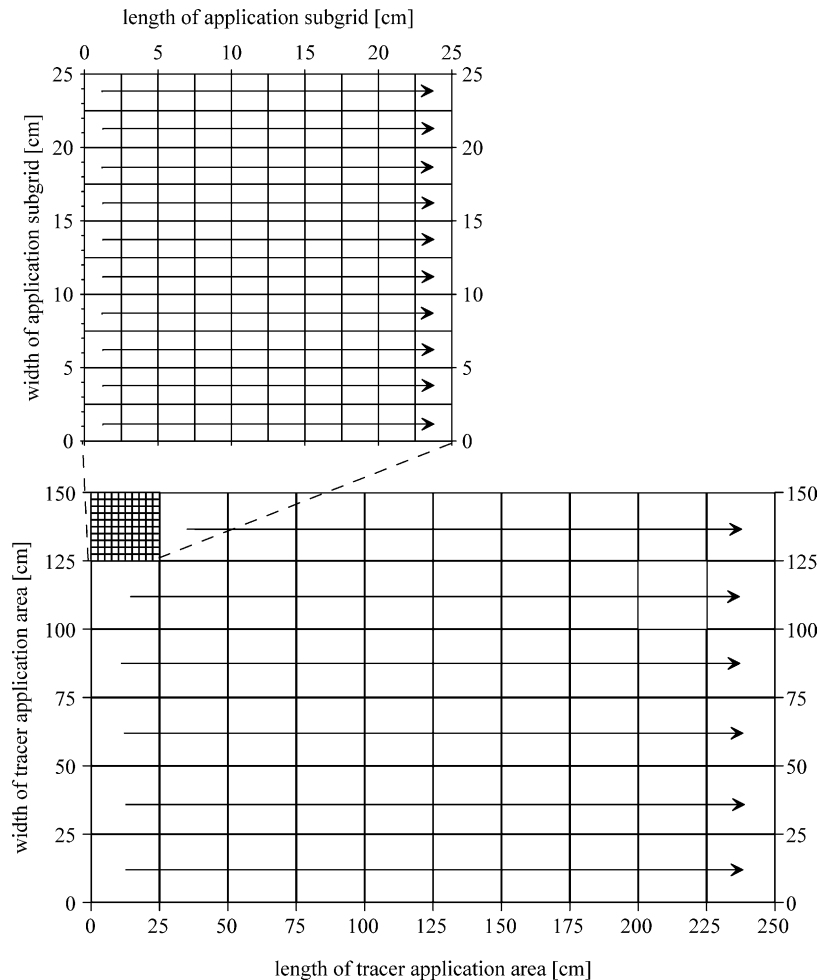


Fig. 5. Plan-view on the tracer application scheme. The mine soil surface was divided into 60 grids of 25×25 cm (bottom), each of which was further subdivided in 100 subgrids of 2.5×2.5 cm (top part magnified). The arrows indicate the sequence of the tracer application procedure within each grid and for the total plot, respectively.

Beginning on September 21st 2001, the soil monolith above the cell-lysimeter was completely sampled to obtain the 3D spatial distribution of residual bromide and to determine physical and chemical soil properties. The soil monolith was divided into 45 columns each located directly above a lysimeter cell and corresponding to its size (i.e., 27×27 cm). The soil columns were further subdivided into 5 vertical sections of 27, 27, 20, 20, and about 16 cm thickness from bottom to top, respectively, resulting in a total of 225 soil blocks. The upper two sections comprised the ameliorated topsoil layer. The thickness of the topsoil segment was changing according to

the local variations in the surface micro relief. The lower three sections comprised most of the original overburden sediments in the inclined dumping structures (Fig. 1). Soil sampling was carried out starting from the adjacent soil pit (Fig. 2b). First, the uppermost row of the front side was sampled by manually cutting the nine block samples one-by-one out of the soil monolith. The complete vertical cross-section followed by working vertically downwards row by row down to the fifth row at the bottom. Afterwards, the next uppermost row of the second cross-section was sampled and so forth towards the bottom row of the fifth vertical cross-section.

Table 1

Cumulative throughfall, suction-cell drainage. Non-capillary gravity drainage, and the ratio between total drainage and throughfall for the total period of the tracer experiment and separated for spring and autumn periods

Period	(a) Through- fall (mm)	(b) Suc- tion-cell drainage (mm)	(c) Non- capillary gravity drainage (mm)	Ratio of (b + c)/ a (–)
7.11.00–21.9.01	211.1	29.6	2.0	0.15
Spring 2001	83.9	22.6	1.3	0.29
Autumn 2001	127.2	7.0	0.7	0.06

In the field, each soil block sample was weighed, and the soil was sieved to obtain fractions of <2, 2–7, and >7 mm in size. Both larger size fractions consisted mainly of lignitic fragments. The size fraction <2 mm was oven-dried (105 °C) for 24 h and weighed. For each of the 225 mine soil samples, a subsample of 20 g from the homogenized <2 mm fraction was suspended in 50 ml of distilled water. After 1 h of shaking at 250 rpm and 10 min of centrifuging at 3000 rpm, the supernatant solution was membrane filtered (0.45 µm) and analysed ion-chromatographically (Dionex 500 IC) for residual bromide. The extraordinarily high sulphate concentrations of this mine soil of up to 23.3 g SO₄^{2–} kg^{–1} dry weight (Hangen, 2003) did not affect bromide

analysis, since the two ions occur at distinctively different positions on the chromatographical spectra.

3. Results and discussion

3.1. Infiltration and drainage

The total amount of water drained from the cell-lysimeter during the tracer experiment (7th of November 2000 till 21st of September 2001) of 31.6 mm includes both the gravity-driven drainage from the perforated pipes (6%) and the suction-driven drainage from the funnel-type cells (94%) (Table 1). Drainage occurred only during two periods in spring and autumn. Most of the drainage (i.e., 76%) was collected during period A (4th of January 2001 till 20th of April 2001). Period A was characterised by several cycles of rain-on-snow events and pronounced snow-melt (Fig. 6). During period B (10th till 21st of September 2001), drainage of 8 mm (i.e., 24%) was triggered by convective rain storms. Drainage amounted to 29% of throughfall during period A and 6% during period B (Table 1). Whereas a maximum of about 71% of the total cross-sectional area (i.e., fractional number of contributing cells) of the lysimeter participated in drainage during period A (Fig. 7, left), this fraction was only 35% during period B (Fig. 7, right). The fractional drainage area increased

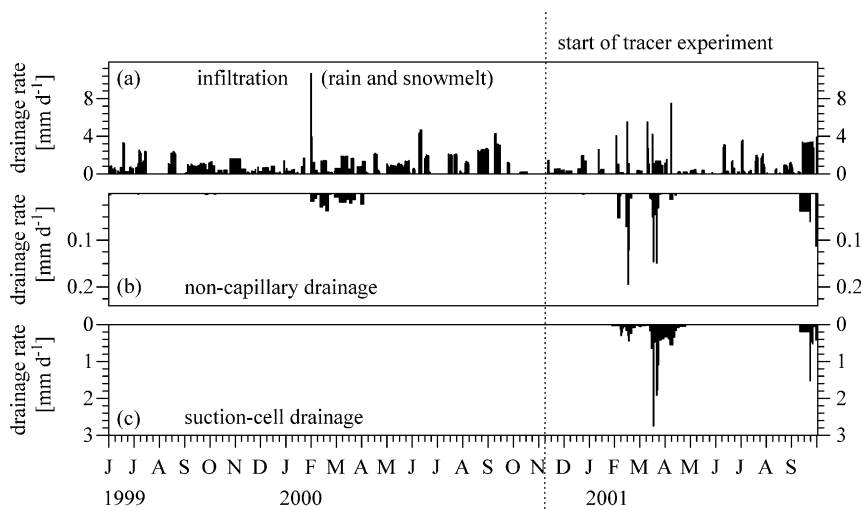


Fig. 6. Soil block-averaged infiltration, non-capillary drainage and suction-cell drainage over the total observation period (June 1999 to September 2001). The dotted line marks the start of the tracer experiment (7th of November 2000).

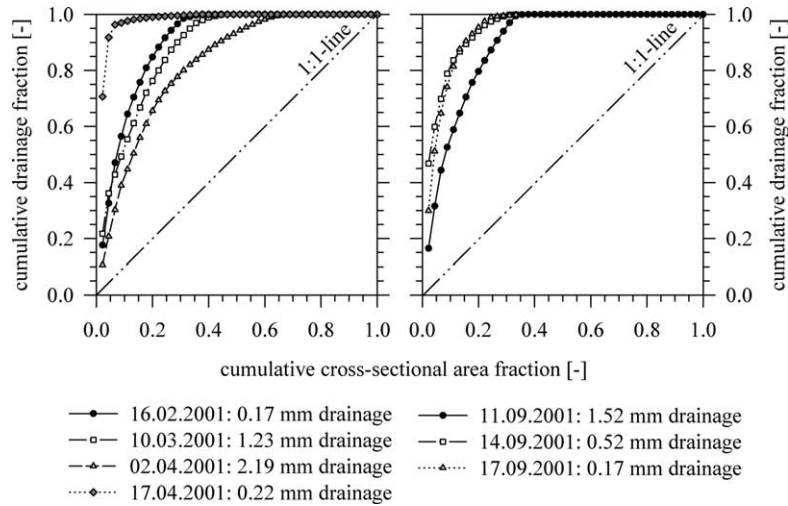


Fig. 7. Mean cumulative drainage at selected sampling dates (increasing order) as related to the contributing cross-sectional drainage area of the total cell-lysimeter for (a) the spring drainage period (4.1.2001–20.4.2001) and (b) the autumn drainage period (1.9.2001–21.9.2001). Drainage depths are given for each sampling date.

with increasing amount of drainage flux. In particular during period A (Fig. 7, left), the fractional area increased up to 70% on April 2nd 2001. During period B, the maximum amount of drainage occurred in the first event on September 11th 2001 (Fig. 7, right). The fractional drainage area did not change as much as a period A, indicating in different wetting pattern in the autumn as compared to the spring.

3.2. Bromide transport

Bromide was found in the gravity as well as in the suction-cell drainage. Cumulative drainage and bromide mass leaching was much larger for suction-cell than for gravity drainage (Fig. 8). The results in Fig. 8 show also that relatively more bromide was exported at the end of the more intensive drainage period A in

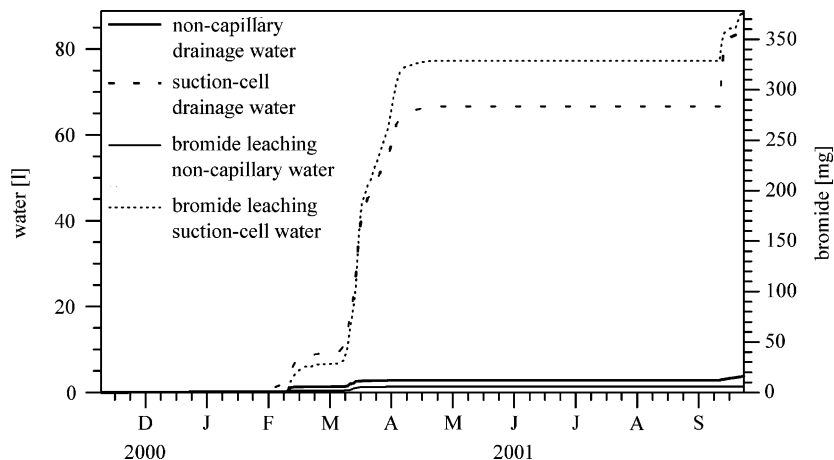


Fig. 8. Cumulative amount of total suction-cell and non-capillary drainage waters (litre) and cumulative amount of bromide leaching with suction-cell and non-capillary water during the tracer experiment period (7.11.2000–21.9.2001).

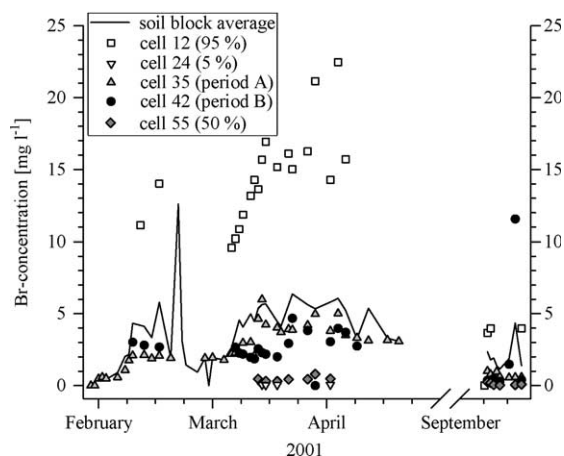


Fig. 9. Bromide concentrations of selected lysimeter cells and the soil block average, respectively, over the leaching period (January–September 2001).

spring 2001. The relatively early occurrence of bromide in the drainage water (i.e., 83 days after application) suggests some kind of preferential flow. However, the findings from this mine soil are not comparable with the bypass-type of preferential flow reported from macroporous soils. While the latter case is characterised by highest leaching and effluent concentrations occurring initially (e.g., Wichtmann et al., 1998; Fortin et al., 2002), bromide concentrations here are gradually increasing to remain at a fairly constant level (Fig. 9).

The mean bromide concentration of suction-cell drainage was higher (4.25 mg l^{-1}) than that of gravity drainage (1.55 mg l^{-1}). Note that mean concentrations were calculated by dividing cumulative leached bromide mass by corresponding drained water volume. During period A, the bromide concentrations were gradually increasing (Fig. 8) and higher than during period B for both drainage fractions. Both drainage periods showed different bromide concentration patterns. The diffusive and dispersive spreading of the tracer during the summer months may have offered access to more immobile pore regions and potential sorption sites, which could result in higher tracer retention (c.f. Jene, 1998). The uptake of tracer by pine trees could also have contributed to a decrease in bromide concentrations. For instance, on an agriculturally used silty loam, 47% of the applied bromide were recovered in the biomass of the cropped corn (Sticksel et al., 1996).

In the following, the soil block average bromide concentration was compared with that of individual lysimeter cells, which were selected according to their cumulative bromide effluent. All 45 cells were ranked according to the amount of bromide leached over the total observation period. The 95-percentile cell (cell 12) represents the upper and the 5-percentile cell (cell 24) the lower part of the spectra (de Rooij, 1996). The 50-percentile or median cell (cell 55), demonstrates the central value. The cells with maximum drainage values serve as a reference, i.e., cell 35 for period A and cell 42 for period B, respectively. When comparing the bromide concentration of selected individual lysimeter cells with that of the soil block average, the magnitude of the spatial variability of bromide transport becomes evident (Fig. 9). While cell 12 showed markedly elevated bromide concentrations of up to 23 mg l^{-1} , those of cell 35 resembled that of the drainage-weighted soil block average of about 4 mg l^{-1} , particularly during drainage period A. In contrast, during period B, bromide concentrations of cell 42 were better corresponding with those of the soil block average. Note that in the time window around the peak concentration of the soil block average (Fig. 9), the five selected cells did not have any drainage water, and thus no bromide concentration values.

To further characterize the spatio-temporal patterns of cumulative bromide leaching from individual lysimeter cells, the 95-, 90-, 50-, 10- and 5-percentile cells, respectively (c.f. de Rooij, 1996), were depicted as a function of cumulative drainage (Fig. 10). Additionally, the cell with maximum drainage served as a reference. Notice the different scales used in Fig. 10. Values ranged between 70 mg bromide (95-percentile) and 0.2 mg bromide (5-percentile). The variation of bromide leaching between cells was more than two orders of magnitude for the total period. In Fig. 10, leached bromide of the 95-percentile cell (i.e., cell 12) was transported by a relatively small amount of drainage water. At the 90-percentile cell 15, the reference cell 35, and the median cell 55, respectively, bromide leaching was somewhat corresponding with the cumulative drainage. The latter case may suggest predominantly convective bromide transport patterns (de Rooij, 1996). The 90-percentile cell 15 and the 50-percentile cell 55 display an inflection point at 36 and 12 mm of drainage,

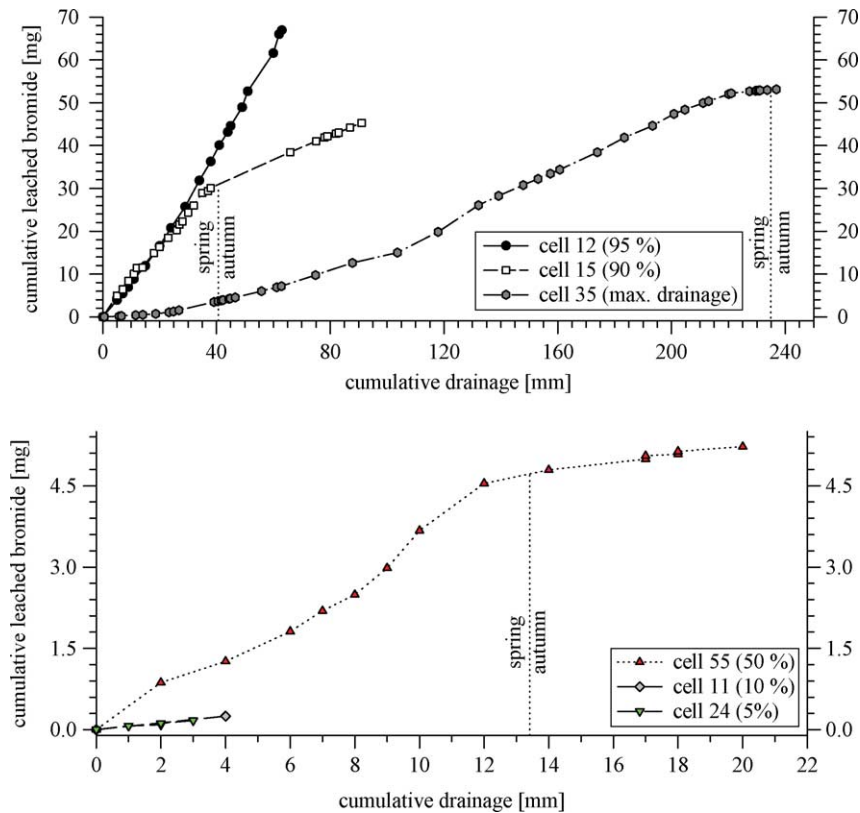


Fig. 10. Cumulative leached bromide of the 95- and 90-percentile lysimeter cell and the lysimeter cell with maximum drainage, respectively (top), and the 50-, 10-, and 5-percentile lysimeter cell (bottom) as a function of cumulative suction-cell drainage.

respectively. These points indicate the transition from drainage period A to period B (Fig. 10). The decrease in bromide effluent concentration may be explained by spreading and uptake of bromide as discussed above (Fig. 8).

The leaching of bromide displayed a distinct non-uniformity (Fig. 11). During period A, 20% of the leached bromide was exported by a single lysimeter cell, which equals approximately 2% of the area fraction. Two cells (approximately 4%) accounted for 36% of the exported bromide. About 30% of the drainage area did not participate in bromide export during period A (Fig. 11). During period B, 67% of the drainage area was excluded from bromide export, while 30% of the leached bromide was delivered by a single lysimeter cell, that corresponds to about 2% of the area fraction. Another 52% of the exported bromide could be attributed to just two cells or about 4% of the area.

The 2D-horizontal distribution patterns of through-fall seem to be less heterogeneous than that of drainage (Fig. 12a and b). During the period A, higher amounts of throughfall were observed at

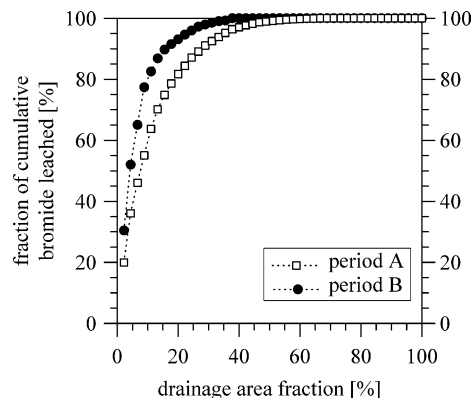


Fig. 11. Cumulative fraction of exported bromide load (decreasing order) as related to the contributing cross-sectional drainage area of the total cell-lysimeter for drainage periods A and B.

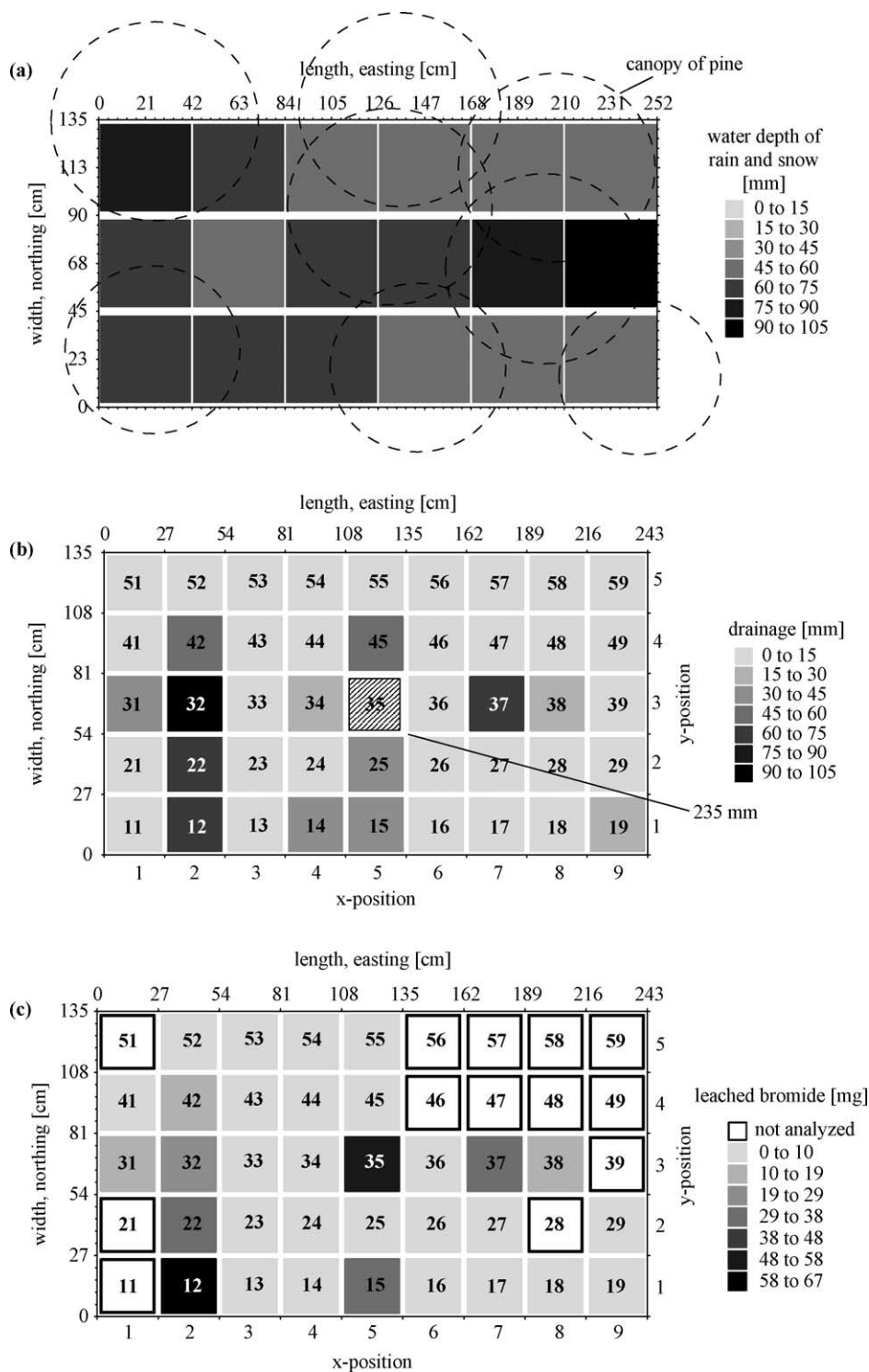


Fig. 12. 2D-horizontal distribution of: (a) cumulative throughfall; (b) drainage; (c) bromide load at 'Bärenbrück' during the spring period (4.1.2001–20.4.2001).

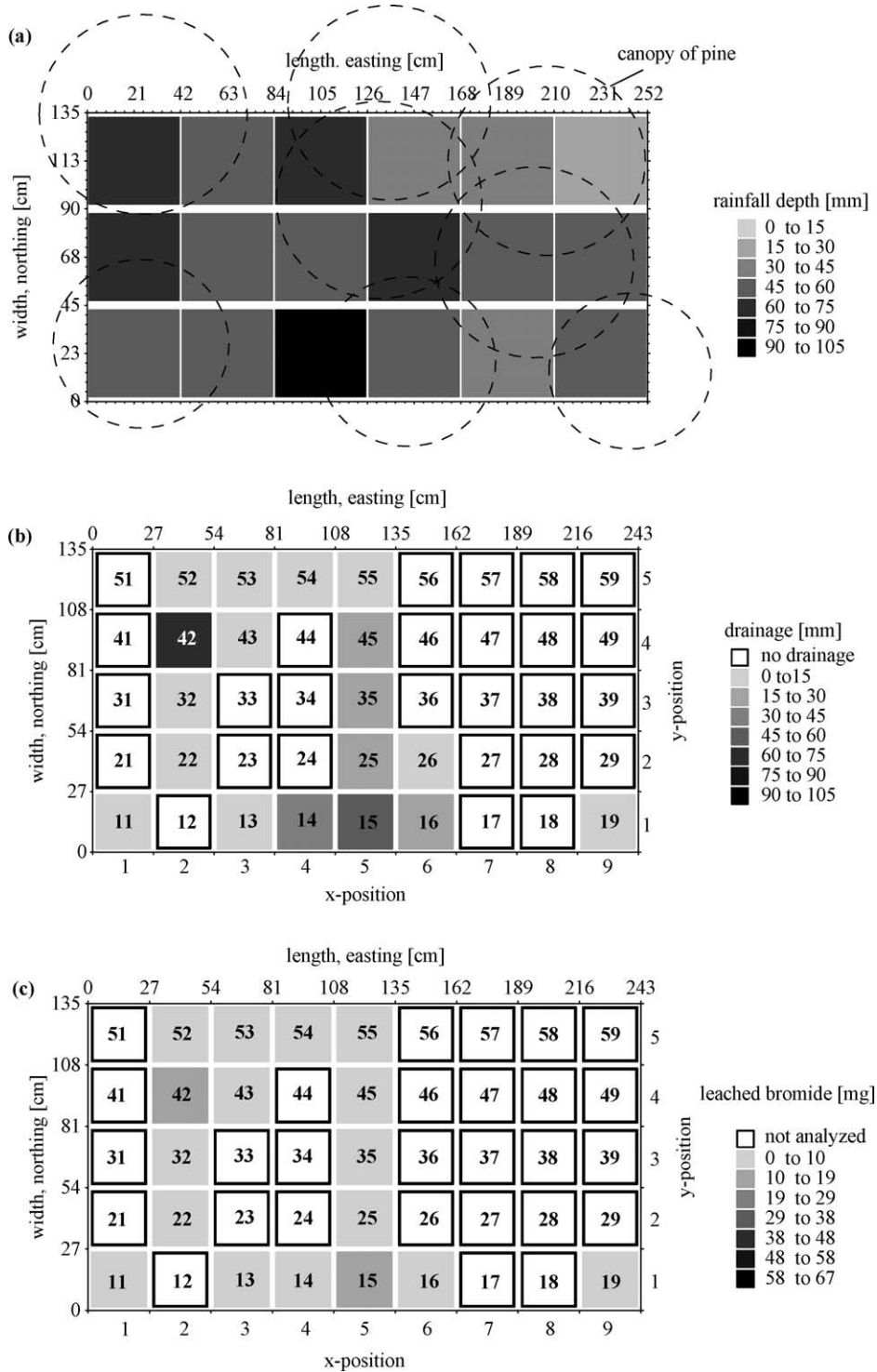


Fig. 13. 2D-horizontal distribution of: (a) cumulative throughfall; (b) drainage; (c) bromide load at 'Bärenbrück' during the autumn period (1.9.2001–21.9.2001).

the left-hand and the central right-hand side of the experimental plot. In contrast, drainage predominantly occurred through central cells 32, 35, 37, and sections at the left front (Fig. 12a and b). The amount of drainage of cell 35 even exceeded that of the accumulated throughfall. The 2D-horizontal distribution of cumulative leached bromide of period A shows inversely (e.g., cells 12, 32, and 34) as well as congruently arranged (e.g., cells 35, 37, and 38) leaching patterns as compared with the corresponding amount of drainage (Fig. 12b and c). Note that for some cells, bromide concentrations of drainage samples could not be analysed because the water volumes collected were smaller than 10 ml and therefore insufficient for duplicate chemical analysis.

During period B, throughfall was more evenly distributed than before with a maximum in the front central left section. Drainage occurred in the central front sections and at a single cell in the upper left section (Fig. 13a and b) indicating a different spatial distribution compared to period A. Again, the distribution of drainage does not correspond with that of throughfall. Most drainage was observed along cell rows 2 and 5 in y-direction (Fig. 13b). Main drainage shifted from the central cells during period A (Fig. 12b) to the central front and the upper left section (cells 15 and 42) during period B (Fig. 13b) largely corresponding with spatial patterns of bromide leaching (Figs. 12c and 13c). The significance of cell 35 and 42, respectively, for the block-averaged bromide concentration of drainage of period A and B, respectively, was demonstrated in Fig. 9. The results clearly indicate that the distribution of drainage does not correspond with that of throughfall horizontally for both periods A and B owing to the horizontal heterogeneity of the aboveground biomass, the root system, the mine soil structures, and the lignitic fragment distributions.

During the 5 months without drainage between April and September 2001, centres of elevated drainage shifted from cell 35 to cell 15 and from cell 32 to 42 by about 27 to 54 cm (Figs. 12b and 13b). Furthermore, we noticed that comparable shifts in the location of drainage centres could be observed even within daily sampling intervals (not shown here). In this mine soil, the shifting of flow paths and patterns is in contrast to some studies on water repellent sandy soil under grass (e.g., Ritsema et al., 1998), where

time-invariable flow paths were observed. On the other hand, periodic lateral shifts of moist soil sections (de Rooij, 1996) and spatially variable centres of drainage (Jene, 1998) were reported from lysimeter studies on sandy and loamy soils. The reasons for the spatially shifting flow paths here must be related to the distribution of antecedent soil moisture and soil chemical conditions, which determine the wettability and permeability conditions following the small-scale spatial distribution of mine overburden sediments. The mine soil in the initial stage of soil development possibly had not yet established stable structures with respect to root system and salt leaching. The configuration of preferential flow regions may easily be changing between individual rainfall events and even during one single event. The latter case was demonstrated in a field study on a drained silty loam with different subsequently applied tracers (Kung et al., 2000). In the field study presented here, periodic shifts of spatial leaching patterns as well as the deviations between throughfall and drainage patterns may probably be affected by varying local distributions of water repellence (Gerke et al., 2001b). The heterogeneously distributed root system of the pine trees (Hangen et al., 2001b) could have affected leaching patterns by creating additional spatial variability of soil moisture that may differ between the beginning and the end of the vegetation period.

3.3. Comparisons between tracers

The spatial leaching patterns of the applied tracers bromide, deuterium, and terbuthylazine are compared for the drainage event on April 2nd 2001 (Fig. 14), when the number of draining cells was highest. Note that the mass scales relative to the differing absolute values were equally classified. Terbuthylazine concentrations in the drainage water of 12 lysimeter cells were below the analytical detection limit of $1 \mu\text{g l}^{-1}$. Only cells 37 and 38 show correspondingly increased bromide and terbuthylazine leaching, while for most cells, bromide and terbuthylazine spatial leaching patterns differed markedly at this date (e.g., cell 12 or 43) (Fig. 14a and c). When comparing the spatial leaching patterns of bromide and deuterium, the number of congruent cells is relatively high. For instance, cells 12, 15, 32, 37, and 38 display increased

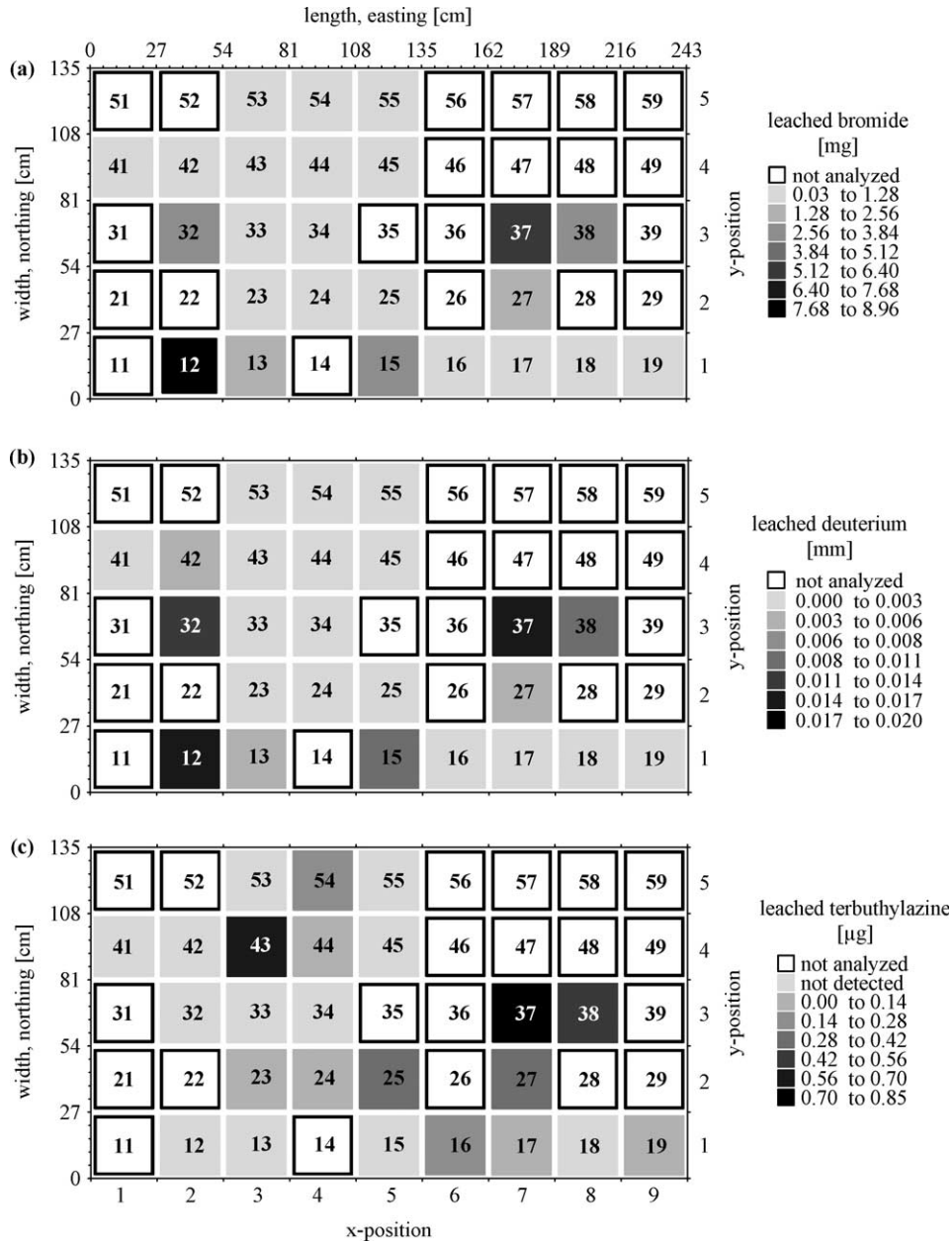


Fig. 14. 2D-horizontal distribution of leached tracers on the 2nd of April 2001: (a) bromide; (b) deuterium; (c) terbutylazine.

bromide as well as deuterium leaching on the 2nd of April 2001 (Fig. 14a and b).

To allow for the comparison of the temporal transport behaviours of bromide, deuterium and terbutylazine the measured values are compiled in Table 2.

The temporal leaching patterns of the three tracers are exemplified for the cells 32 and 35 (Fig. 15), which had contrasting spatial patterns (Fig. 14). The first increase of the relative bromide concentrations in January 2001 to values of up to 0.0007 corresponded with the first appearance of relevant amounts of

Table 2

Ranges of suction-cell drainage's absolute and relative concentrations of bromide, deuterium, and terbuthylazine for those sampling days with all three applied tracers analysed

Date	Bromide				Deuterium				Terbuthylazine			
	Absolute (mg/l)		Relative (10^{-3})		Absolute (‰)		Relative (10^{-3})		Absolute ($\mu\text{g/l}$)		Relative (10^{-3})	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
07.02.01	1.08	3.91	0.35	1.25	1553.5	7431.7	0.44	1.46	<1	<1	–	–
14.02.01	1.69	5.95	0.54	1.90	2361.5	9259.8	0.58	1.78	<1	1.9	–	0.03
01.03.01 ^a	0	0	0	0	1169.7	1169.7	0.38	0.38	1.5	1.5	0.02	0.02
02.03.01 ^a	1.95	1.95	0.62	0.62	1533.3	1533.3	0.44	0.44	1.9	1.9	0.03	0.03
05.03.01 ^a	1.79	1.79	0.57	0.57	2169.6	2169.6	0.55	0.55	<1	<1	–	–
07.03.01	0.59	9.59	0.19	3.07	1301.0	5118.8	0.47	1.06	<1	<1	–	–
20.03.01	3.39	7.71	1.08	2.47	3371.5	9017.4	0.76	1.73	<1	1.2	–	0.02
02.04.01	0	14.28	0	4.57	654.6	17,349.9	0.29	3.18	<1	2.3	–	0.04
10.09.01	0.35	2.92	0.11	0.93	168.5	264.5	0.20	0.22	<1	<1	–	–
21.09.01	0.40	1.90	0.13	0.61	202.5	263.5	0.21	0.22	<1	<1	–	–

^a Only one water sample retrieved on this date.

drainage water (Fig. 6). A second increase in relative concentrations to maximum values of about 0.0015 occurred between mid-March (cell 32) and early April 2001 (cell 35) when drainage rates were higher (Fig. 15). In September 2001, relative bromide concentrations were generally lower at values of about 0.0002. The relative concentrations of deuterium were slightly higher at cell 32 and lower at cell 35

as compared to those of bromide, but both tracers showed mostly similar and corresponding temporal behaviour in effluent concentrations (Fig. 15). Also for the autumn drainage period B, both relative tracer concentrations were in the same order of magnitude. Similar shapes of the breakthrough curves for bromide and deuterium were also observed by Bowman (1984) and Stamm et al. (2002), however,

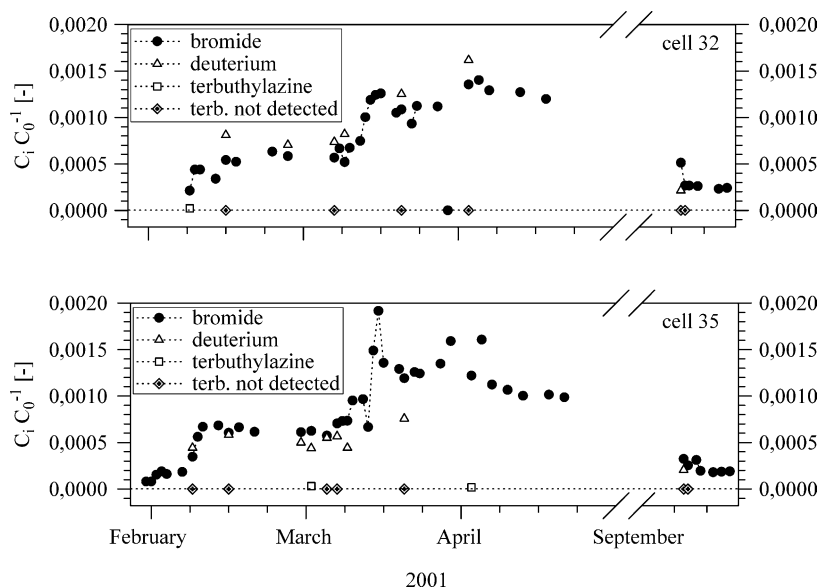


Fig. 15. Relative concentrations of bromide, terbuthylazine, and deuterium in the drainage waters collected from lysimeter cells 32 (top) and 35 (bottom) during spring and autumn drainage periods.

the relative bromide concentrations were higher than those of deuterium. A relative acceleration of bromide compared to deuterium resulting from anion exclusion (Russow et al., 1996) is less effective in coarse-textured rather than in fine-textured soils of high charge density (Bowman, 1984). An almost identical breakthrough of bromide and deuterium was found in lysimeter experiments with undisturbed soil columns from the opencast mine ‘Espenhain’ located in the Central German lignite mining district (Knappe et al., 2001). Similar to the mine soil from ‘Bärenbrück’ of this study, the soil from ‘Espenhain’ was coarse-textured, contained lignitic components, and had highly acidic drainage water with pH-values down to 2 due to ongoing pyrite oxidation. A possible sorption of bromide at low pH-values at protonated surfaces (Russow et al., 2001) of mineral or lignitic mine soil compounds may counteract an accelerating effect of anion exclusion on bromide transport. In total, both factors could possibly result in an apparently ‘ideal’ tracer behaviour as suggested by Knappe et al. (2001).

The sorptive chemical terbuthylazine was found only during the first drainage period at low relative concentrations in drainage waters of cells 32 and 35 (Fig. 15). Detectable levels of terbuthylazine concentrations were found in the drain effluent of only 17 lysimeter cells (not shown here). For cell 32, terbuthylazine was found in water samples collected in February 2001. For cell 35, terbuthylazine was found in samples from March and April 2001.

Levels of relative terbuthylazine concentrations were about two orders of magnitude lower than those of bromide and deuterium (Fig. 15). Terbuthylazine leaching found in 7 out of 16 lysimeter cells at a site with loamy sand was attributed to preferential flow (Trapp et al., 1995). The detection of terbuthylazine, although at low concentrations, in the drainage water from ‘Bärenbrück’ suggests that bypass-type preferential flow may occur under intensive snow melt at this lignitic mine soil. During high infiltration at moist soil conditions, there seems to be a reasonable leaching risk even of sorptive contaminants. However, the relatively few findings and small concentrations of terbuthylazine indicate that bypass-type preferential flow is not a dominating transport mechanism, as already discussed above for bromide leaching (Fig. 8).

3.4. 2D-vertical distribution of residual bromide

An integrated 2D-vertical distribution of residual bromide content in the particle-size fraction <2 mm of the mine soil was obtained by summing up the bromide mass of the block samples over the width (5 blocks in y-direction) of the mine soil monolithe. The 2D-vertical distribution of bromide content indicates higher residual values in the topsoil, especially at the right-hand-side (Fig. 16, top). In the subsoil, higher bromide contents can be observed at a few spots that appear to be located along inclined structures and to correspond with the location of

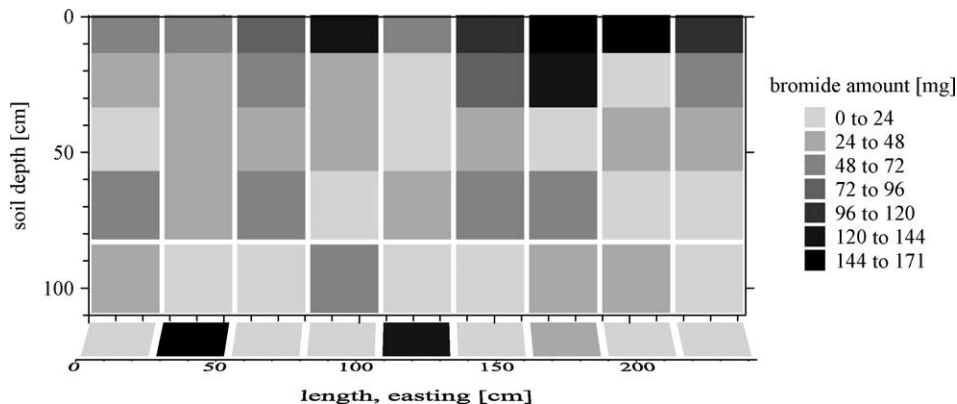


Fig. 16. 2D-Vertical distribution of residual bromide in the mine soil for the particle size fraction <2 mm above the cell-lysimeter and bromide content in cell effluent leached over the experimental period. Grid-block values were integrated over the width (y-direction) of the mine soil block.

suction cells with highest bromide leaching by drainage (Fig. 16, bottom). Inclined soil structural patterns can immediately be observed by visual inspection of the mine soil profile (Fig. 1). The inclined layer with the brighter soil material (Fig. 1) presumably diverted vertically moving soil water laterally towards the fifth cell row thereby inducing a disproportionately high amount of drainage and bromide leaching at this row (see Figs. 12 and 13). This process of flow concentration appears to be similar to ‘funnelling’ (Kung, 1990). Another diagonally oriented bright structure points to the right hand side outward of the soil block (Fig. 1). Probably, bromide labelled water was diverted out of the studied soil volume causing low water and bromide leaching at the right hand side (Figs. 12 and 13). The observations suggest that inclined spoil ripple structures from dumping of overburden have a dominating effect on the spatial distribution of solute movement in the lignitic mine soil.

It was beyond the scope of this study to separate the residual bromide content in adsorbed and dissolved fractions since bromide distribution coefficients for the various lignitic and mineral components of different degree of degradation and acidification were not easily available.

3.5. Quantitative summarizing analysis

The cell lysimeter and multitracer study showed that drainage and leaching patterns in this lignitic forest-reclaimed mine soil are heterogeneous in both space and time. While snow melt-induced drainage amounted to 28% of the corresponding throughfall depth, only 6% of throughfall were exported during convective rainfall after the vegetation period.

Over the total observation period from November 2000 till September 2001, gravity drainage amounted to 2 mm, which equals about 7% of assessed suction drainage (29.6 mm). The bromide concentration of suction drainage was almost threefold of that of gravity drainage. This points to different pore regions, that are hydraulically active during gravity and suction drainage, respectively. Overall, drainage was restricted to 32 out of a total of 45 cells, i.e., 71%, in spring and to 16 cells in autumn, respectively, equalling 35% of the total cross-sectional lysimeter catch area. A seasonal shift of drainage centres of up

to 54 cm became evident, that followed single spoil ripple structures. Dynamically changing leaching patterns indicate that the development of preferential flow paths depends on the antecedent soil moisture content and the infiltration regime. Except individual cells, drainage water export corresponded mostly to leached bromide and deuterium loads, while exported loads of the sorptive terbuthylazine significantly deviated from the amount of drainage. The observed spatio-temporal leaching behaviour of bromide largely resembled that of deuterium. Despite the mostly acidic conditions in the subsoil, bromide was found to be transported similarly to deuterium and was proved to be a useful tracer.

Preferential flow paths are probably affected by the small-scale heterogeneity of lignitic and mineral components that may be responsible for water repellence. Moreover, larger structures such as tree root systems, which seasonally cause spatially heterogeneous soil water uptake, as well as technogenic spoil ripples forming inclined layers influence the configuration of preferential flow paths.

The ‘funnelling’-type of preferential flow seems to be the predominant mechanism in the lignitic mine soil given a drainage depth of a single individual cell that reached up to 125% of the corresponding infiltration depth. This ‘funnelling’ is probably induced by inclined spoil ripple structures, which result from overburden dumping and sediment heterogeneity together with water repellent zones.

4. Conclusions

The cell lysimeter and tracer study showed that drainage and leaching patterns in this lignitic forest-reclaimed mine soil are heterogeneous in both space and time. Most lysimeter cells did not yield any drainage over the total experimental period. Drainage was restricted to a few out of a total of 45 cells. Except individual cells, drainage water export predominantly corresponded mostly to leached bromide and deuterium loads, while terbuthylazine significantly deviated. Despite the mostly acidic conditions in the subsoil, bromide was found to be transported relatively similar to deuterium.

The spatial distributions of drainage and effluent concentrations indicate that for this mine soil several

different mechanisms may be involved in flow and transport that each depend in a complex way on inclined structures, on heterogeneous and varying soil properties, and on initial and boundary conditions. Gravity drainage and early appearance of tracers in drainage water indicate that bypass-type preferential flow may occur at this site. However, bypass flow does not seem to be quantitatively important, since the reactive tracer terbutylazine appeared in the drainage water only at relatively low concentrations during the first drainage period and the amount of gravity drainage was small compared to suction-cell drainage.

Due to the spatial variability of drainage and solute leaching, estimated water and element budgets of mine soils that rely on small-volume or point observations have to be interpreted cautiously. The leaching patterns suggest that the ‘funnelling’-type of preferential flow may be regarded as the predominant mechanism although fingering and bypass flow may also occur at the reclaimed mine soil.

Leaching of sorptive pesticides and contaminants at mine soil sites can not be excluded under intensive snow melt conditions and convective summer rainstorms. Hence, an effective monitoring should focus on the linear structures, where two adjacent spoil ripples intersect, because here the bulk of leached contaminants can be expected. Adequate simulations of contaminant leaching in afforested mine soils might also require an appropriate representation of the specific anisotropic spoil ripple structures.

Depending on the respective initial and boundary conditions, a complex of different types of preferential flow may occur during initial soil development of relatively ‘young’ mine soils compared to ‘naturally’ developed soils, where local equilibrium conditions and characteristic patterns have already been established.

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